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Unit 4 Collaborative Wiki Development: Security Frameworks

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Introduction

Security frameworks provide structured approaches to managing cyber risks across different industries. This essay briefly evaluates suitable frameworks for an international bank, a large hospital, and a food manufacturing factory, along with key testing and implementation recommendations (Kirvan, 2021).

International Bank

An international bank requires strong governance and compliance due to high financial risk and regulatory pressure. The **NIST Cyber Security Framework (CSF)** supports structured risk management, while **ISO/IEC 27001** provides an Information Security Management System (ISMS) for formal control of security processes. **PCI DSS** is essential for protecting payment card data, and **CIS Controls** provide practical security baselines. **MITRE ATT&CK** can also be used to understand attacker behaviour and improve detection strategies (MITRE, 2024).

Large Hospital

Hospitals require frameworks that prioritise patient safety and data protection. **ISO/IEC 27001** supports secure handling of patient data, while **NIST CSF** provides a structured approach to risk management. Healthcare regulations such as **HIPAA** (or regional equivalents) ensure legal compliance. **CIS Controls** help secure legacy and endpoint systems, and **MITRE ATT&CK** is useful for analysing ransomware threats commonly targeting healthcare environments (Kirvan, 2021).

Large Food Manufacturing Factory

Food manufacturing environments require both cybersecurity and operational safety controls. **ISO 22000** focuses on food safety, while **IEC 62443** is designed for securing industrial control systems (ICS/SCADA). **ISO/IEC 27001** supports information security governance, and **NIST CSF** provides overall cyber risk management. **CIS Controls** offer practical implementation guidance for baseline security (ISO, 2022; IEC, 2018).

Testing and Recommendations

International Bank

Banks should conduct regular penetration testing of online banking systems and APIs, enforce multi-factor authentication (MFA), and continuously monitor for fraud and anomalies. Regular compliance audits (ISO 27001 and PCI DSS) and phishing awareness training are also essential. Encryption of sensitive data should be enforced at all stages.

Large Hospital

Hospitals should perform regular vulnerability assessments of clinical systems, segment networks between clinical and administrative environments, and ensure robust backup and disaster recovery testing. Role-based access control (least privilege) and incident response planning for ransomware attacks are also critical.

Large Food Manufacturing Factory

Manufacturers should test ICS/SCADA systems for vulnerabilities, enforce IT/OT network segmentation, and monitor production systems for anomalies. Supplier security assessments and combined cybersecurity-safety audits should also be carried out. Resilience testing for system downtime is essential.

Conclusion

Overall, effective security depends on combining governance frameworks such as ISO/IEC 27001 and NIST CSF with industry-specific standards like PCI DSS, HIPAA, ISO 22000, and IEC 62443. When combined with technical controls and threat intelligence frameworks such as MITRE ATT&CK, organisations can improve resilience and incident response capability (Kirvan, 2021).

References

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